Lecture 18: Review

$T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ linear transf.

$$T\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}1\\2\\1\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = \begin{bmatrix}-2\\-4\\-2\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = \begin{bmatrix}4\\9\\5\end{bmatrix}$$

$$T\left(\begin{bmatrix}1\\2\\3\end{bmatrix}\right) = \begin{bmatrix}1 & -2 & 4\\2 & -4 & 9\\1 & -2 & 5\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$$

$T(v) = Av \implies \text{range}(T) = \text{col}(A)$

$\dim \text{range}(T) = \text{rank}(A)$
$= \#\text{ pivots}$
$= \underline{\underline{2}}$

$$A \xrightarrow{\text{Elim}} \begin{bmatrix}1 & -2 & 4\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix} = U$$

$\text{kernel}(T) = \{x \in \mathbb{R}^3 : T(x) = 0\} = \text{null}(A)$.
↳ has dimension $3-2=1$.

T is not onto, not one-to-one, so not invertible.

$\det A = \det U = 0$.

$$B = \begin{bmatrix}1 & -2 & 4\\2 & -4 & 9\\1 & -1 & 5\end{bmatrix}$$

$$\begin{aligned}\det B &= 1\cdot(-4)\cdot 5 + (-2)\cdot 9\cdot 1 + 2\cdot(-1)\cdot 4\\ &\quad - 4\cdot(-4)\cdot 1 - 9\cdot(-1)\cdot 1 - (-2)\cdot 2\cdot 5\\ &= \cancel{-20} - 18 - 8 + 16 + 9 \cancel{+20}\\ &= -26 + 25 = -1\end{aligned}$$

$$\left[\begin{array}{ccc|ccc}1 & -2 & 4 & 1 & 0 & 0\\2 & -4 & 9 & 0 & 1 & 0\\1 & -1 & 5 & 0 & 0 & 1\end{array}\right] \to$$

$$\to \left[\begin{array}{ccc|ccc}1 & -2 & 4 & 1 & 0 & 0\\0 & 0 & 1 & -2 & 1 & 0\\0 & 1 & 1 & -1 & 0 & 1\end{array}\right] \to$$

$$\to \left[\begin{array}{ccc|ccc}1 & -2 & 4 & 1 & 0 & 0\\0 & 1 & 1 & -1 & 0 & 1\\0 & 0 & 1 & -2 & 1 & 0\end{array}\right]$$

$$\to \left[\begin{array}{ccc|ccc}1 & -2 & 0 & 9 & -4 & 0\\0 & 1 & 0 & 1 & -1 & 1\\0 & 0 & 1 & -2 & 1 & 0\end{array}\right]$$

$$\to \left[\begin{array}{ccc|ccc}1 & 0 & 0 & 11 & -6 & 2\\0 & 1 & 0 & 1 & -1 & 1\\0 & 0 & 1 & -2 & 1 & 0\end{array}\right]$$

B^{-1}

$$\det B = 1\cdot\begin{vmatrix}-4 & 9\\ -1 & 5\end{vmatrix} - (-2)\begin{vmatrix}2 & 9\\ 1 & 5\end{vmatrix} + 4\begin{vmatrix}2 & -4\\ 1 & -1\end{vmatrix}$$

$$= 1\cdot(-11) + 2\cdot 1 + 4\cdot 2 = -1 \,//$$

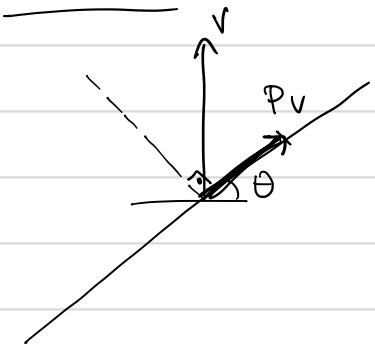
$$C = \begin{bmatrix}1 & 0 & -2 & 4\\ 10 & 4 & 20 & 30\\ 2 & 0 & -4 & 9\\ 1 & 0 & -1 & 5\end{bmatrix}$$

+	−	+	−
−	+	−	+
+	−	+	−
−	+	−	+

Expand along column 2.

$\det C = 4\cdot\det B = -4$

Thus C is invertible.



$$P(v) = T_{+\theta}(P_0(T_{-\theta}(x)))$$

$$T_\theta(x) = Ax$$

$$A = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$$

$$P_0\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}x\\ 0\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}$$

$$\begin{bmatrix}c & -s\\ s & c\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}c & s\\ -s & c\end{bmatrix}$$

$$= \begin{bmatrix}c & s\\ 0 & 0\end{bmatrix}\begin{bmatrix}c & s\\ -s & c\end{bmatrix} = \begin{bmatrix}c^2-s^2 & 2cs\\ 0 & 0\end{bmatrix}$$

$$= \begin{bmatrix}\cos(2\theta) & \sin(2\theta)\\ 0 & 0\end{bmatrix} =: C$$

$P(x) = C\cdot x$ for all x.